

Carbon Dioxide Synergistic Enhancement of Supercritical Methanol on PET Depolymerization for Chemical Recovery

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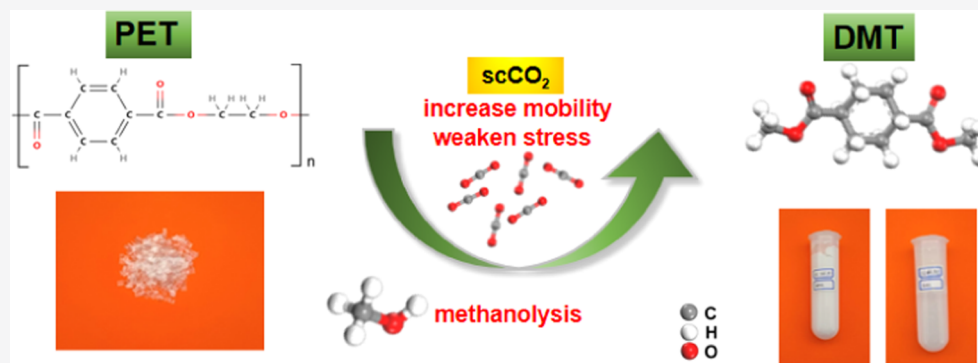


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ABSTRACT: Poly(ethylene terephthalate) (PET) is an important polyester material widely used in beverage bottles, fibers, engineering plastics, and film fields. The typical method of waste recycling is to use supercritical methanol to depolymerize PET into monomers such as dimethyl terephthalate (DMT) and ethylene glycol (EG). However, the supercritical methanol alcoholysis process of PET usually faces the problems of high temperature and pressure and high energy consumption. To further enhance the alcoholysis efficiency, CO₂-enhanced supercritical methanol alcoholysis of PET polyester is used in this work, and the general supercritical methanol alcoholysis process is systematically compared. The research results revealed that the DMT yield of PET polyester by CO₂-enhanced supercritical methanol alcoholysis was maximized under the same reaction conditions, up to 95%. Combined with the *p-x,y* phase diagram, the mechanism of carbon dioxide intensification is discussed, which can provide guidance for the industrial large-scale recycling of PET.

1. INTRODUCTION

Plastics play an important role in promoting the progress of human society, which are widely used in packaging, clothing and sporting equipment, biomedical devices, and electronic components.^{1–5} Production of plastics in the world was 359 million tons in 2019, compared with 81.84 million tons in China. Plastics have become a crucial part of lifestyle by meeting the needs of production and life. However, environmental pollution is also caused by waste plastics, which have been identified as a huge hassle.

About 80% of the polymers are thermoplastics,^{6–9} which are commonly used in food packaging, textile fiber, construction, and coating industries. Typically, it is poly(ethylene terephthalate) (PET), with excellent thermal and mechanical properties, which is a semicrystalline and thermoplastic polyester with high strength, high transparency, and high safety. At present, the treatment of plastic wastes includes landfill, incineration, and resource recovery.

Among these treatment methods, recycling and reusing of waste plastics should be supported and encouraged. There are three types of recycling for plastic wastes:^{10–15} physical

recycling, chemical recycling, and energy recovery. For the recycling of PET, the chemical recycling method is the most desirable process because PET is converted to its monomers, which can then be reused. Chemical recycling processes for PET are divided as follows:^{10–15} hydrolysis, methanolysis, glycolysis, and other processes. The reaction products of hydrolysis are terephthalic acid (TPA) and ethylene glycol (EG), the main products of PET methanolysis are dimethyl terephthalate (DMT) and ethylene glycol (EG), and the products of glycolysis are dihydroxyethyl terephthalate (BHET) and oligomers.

In recent years, the application of supercritical fluid in the treatment of waste plastics is very attractive. Supercritical fluids

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such as water and methanol are excellent reaction media for the depolymerization or decomposition of PET.^{16–20} Using supercritical fluids, depolymerization of PET proceeds rapidly, and it more easily undergoes decomposition into monomers. The disadvantage of the supercritical water hydrolysis process is that the severe reaction conditions (400 °C, 40 MPa) are not conducive to industrialization, and it leads to a low yield of ethylene glycol (10%).^{16–20} In contrast, the conditions for supercritical methanolysis are relatively mild. Takeshi Sako²¹ et al. used pure PET as a raw material and proposed the application of supercritical methanol (300 °C, 9–23.0 MPa) to depolymerize PET into DMT and ethylene glycol, and the yield of DMT was about 80% within 30 min. They found that PET could be completely depolymerized into monomers quickly and completely with supercritical methanol, and the decomposition rate of PET was much higher than that of traditional methanolysis and avoids the use of metal catalysts such as zinc acetate. Kumazawa²² et al. investigated the effect of supercritical methanol alcoholysis of PET into DMT at 280–310 °C, the methanol to PET ratio (weight) of 10, and the reaction time of 10–70 min. It was found that when the temperature was 280 °C and the reaction time was 40 min, the yield of DMT was about 88%, and the yield of DMT was increased to 90% by prolonging the reaction time to 70 min. However, with an elevated temperature of 300 °C and 310 °C and a reaction time of 40 min, the yield of DMT could reach up to 98% and remained basically stable. Goto²³ et al. used pure PET as a raw material and proposed the supercritical methanol alcoholysis of PET to DMT and ethylene glycol at 300 °C and 14.7 MPa; the ratio (weight) of methanol to PET was 5, and the reaction time was 3–60 min. Similarly, the yield of DMT was about 85% with a reaction time of 40 min, and the yield was increased to 95% by continuing to extend the reaction time to 60 min. It was found that when the reaction time was 40 min, the yield of DMT was about 85%, and when the reaction time was increased to 60 min, the yield of DMT was about 95%. According to the above-mentioned literature, the reaction temperature for supercritical methanolysis of PET needs to reach at least 300 °C to achieve monomer yields as high as 98%. However, the methanol to PET ratio (by weight) is small and the reaction time requires at least 60 min. It can be found from the above-mentioned studies that when the yield of monomers is as high as 98%, the reaction temperature of supercritical methanolysis of PET needs to reach at least 300 °C, and when the methanol to PET ratio (weight) is small, it takes at least 60 min. On the premise of obtaining the same yield, when the reaction time is reduced, the methanol to PET ratio (weight) needs to be increased. Mishra²⁴ et al. got the optimal percentage conversion of PET into DMT and EG with the catalysts $\text{ZnC}_4\text{H}_6\text{O}_4$ and $\text{Pb}(\text{C}_2\text{H}_3\text{O}_2)_2$, which was 97.8% (at 120 °C) and 100% (at 130 and 140 °C), respectively. Kurokawa²⁵ et al. increased the yields of DMT and EG up to 88 and 87% by the use of a toluene/methanol mixed solvent in the presence of aluminum Ti isopropoxide (AIP), respectively. Wang²⁶ et al. adopted ultrasmall ZnO nanoparticles dispersed in methanol and ethylene glycol as pseudohomogeneous catalysts for methanolysis of PET. They obtained a PET conversion of almost 97%, and the DMT high yield (95%) during 15 min at 170 °C showed the promising efficiency of this process, with the excellent activity of 553 g PET h⁻¹ (g ZnO). Pudack²⁷ et al. degraded PET into two main reaction products through methanolysis, DMT and EG. The separation through distillation combined with crystallization eliminates

non-PET segments and risky impurities from copolymers, affording monomers of high purity required for repolymerization designs. In the studies reported above, they used high temperature (300 °C) or long reaction time (60 min) or catalysts to obtain a higher yield of DMT. Therefore, to reduce the reaction temperature, reduce the amount of solvent methanol, shorten the reaction time, reduce the energy consumption in the reaction process, improve the reaction efficiency, and realize industrialization better, CO₂ is used to strengthen the supercritical methanolysis of PET.

In this study, CO₂ was used to strengthen the supercritical methanolysis of PET. A series of experiments were conducted to further study the methanolytic depolymerization of PET under the supercritical state of methanol, particularly to determine the effects of the temperature, weight ratio of methanol to PET, and reaction time on the DMT yield. Furthermore, a systematic comparison with the general process of PET supercritical methanol depolymerization was carried out. Also, it was compared with the general process of supercritical methanolysis of PET systematically. It was found that under the same reaction conditions, supercritical methanolysis of PET with CO₂ can effectively improve the yield of monomers. On this basis, the optimum reaction conditions of supercritical methanolysis of PET with CO₂ were obtained.

2. EXPERIMENTAL SECTION

2.1. Materials. The PET used in this study came from a waste mineral water bottle, which was first washed with deionized water at 100 °C,²⁸ broken into flakes, and then dried at 50 °C to remove moisture and volatile compounds. Anhydrous methanol with a purity higher than 99.8% was used as a solvent and purchased from Tianjin kemio Chemical Reagent Co., Ltd. CO₂ with 99% purity was purchased from Dalian Junfeng Gas Chemical Co., LTD.

2.2. Experimental Process. In this study, a batch-type reactor was used for the depolymerization of PET. The experimental apparatus is shown in Figure 1; in the apparatus,

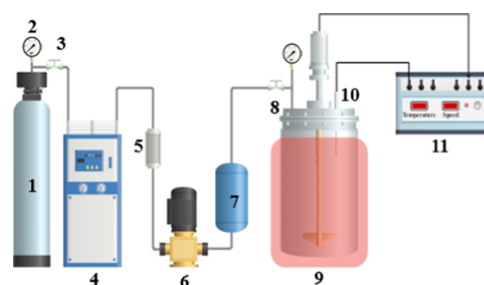


Figure 1. Apparatus for the depolymerization of PET. (1) CO₂ gas cylinder; (2) pressure gauge; (3) needle valve; (4) cooling circulating pump; (5) filter; (6) plunger pump; (7) buffer tank; (8) high-pressure reactor; (9) electric heating furnace; (10) thermocouple; and (11) PID controller.

the high-pressure reactor (8) was the core part of the device, through which a certain amount of PET polyester and 50 mL of methanol were loaded and reacted; the CO₂ gas cylinder (1) is used to provide the required CO₂ for the experiment, and it was disconnected from the reactor through a valve after pressurizing the reactor; and the temperature of the reactor was controlled cooperatively using an electric heating furnace

(9), thermocouple (10), and proportion integration differentiation (PID) controller (11).

2.2.1. Experimental Procedure of Supercritical Methanolysis of PET.

- (1) A total of 3.96–6.60 g of PET after pretreatment and 50 mL of methanol were loaded into the reactor, the initial temperature is room temperature, the reaction temperature was set at 240–270 °C, and the stirring speed was set at 1000 rpm.
- (2) An electric heating furnace was used for heating, the thermocouple is used for temperature measurement, and the PID controller is used to control the temperature and keep the constant temperature at 240–270 °C.
- (3) It took about 40 min for the reactor to reach a given reaction temperature and pressure, and the reactor was kept at the reaction temperature for 20–60 min; then, the reactor was taken out of the electric heating furnace, and it was then cooled rapidly in a water–ice bath to stop the reaction.
- (4) After the reactor vessel was cooled to room temperature, the reaction products were taken out, and the solid-phase products and liquid-phase products were separated by filtration.

2.2.2. Experimental Procedure of CO₂-Enhanced Alcoholysis.

- (1) A total of 3.96–6.60 g of PET after pretreatment and 50 mL of methanol were loaded into the reactor; the CO₂ gas cylinder was used to provide the required CO₂ for the experiment, and it was disconnected from the reactor through a valve after pressurizing the reactor; the initial temperature is room temperature; the reaction temperature was set at 240–270 °C; and the stirring speed was set at 1000 rpm.
- (2) An electric heating furnace was used for heating, the thermocouple was used for temperature measurement, and the PID controller was used to control the temperature and keep the constant temperature at 240–270 °C.
- (3) It took about 40 min for the reactor to reach a given reaction temperature and pressure, and the reactor was kept at the reaction temperature for 20–60 min; then, the reactor was taken out of the electric heating furnace, and it was then cooled rapidly in a water–ice bath to stop the reaction.
- (4) After the reactor vessel was cooled to room temperature, the CO₂ was discharged slowly through a valve, the reaction products were taken out, and the solid-phase products and liquid-phase products were separated by filtration.

2.3. Analysis and Characterization. The solid-phase products were dried to a constant weight at 50 °C; the weight at this time was recorded, and they were ground into a fine powder for high-performance liquid chromatography (HPLC). The operating parameters of liquid chromatography are as follows: a reverse-phase Zorbax-C8 (4.6 mm × 250 mm) column and a SPD-1 ultraviolet detector set at 254 nm were used, 40 mg/L sample solution for analysis was prepared by dissolving a certain amount of the dried solid product in a 80:20 (v/v) methanol/water solution, which was also used as a mobile phase, the injection volume was 5 μL, the column temperature was room temperature, and the column was balanced with the mobile phase for 20 min. The yield of DMT

was represented as the percentage of the DMT quality in the solid-phase products calculated from HPLC data against the theoretical DMT quality. The concentrations of the components in liquid products were analyzed by gas chromatography (GC). The operating parameters of gas chromatography are as follows: a HP-5 capillary column was used, the injection volume was 1 μL, the column temperature was 160 °C, the detector temperature and the inlet temperature were 200 °C.

$$\text{DMT yield (\%)} = \frac{\text{quality of actual output DMT} \times 100}{\text{theoretical DMT quality}} \quad (1)$$

3. RESULTS AND DISCUSSION

3.1. Investigation of the Supercritical Methanolysis of PET. First, to find the best reaction conditions for the supercritical methanolysis of PET, the effects of the weight ratio of methanol to PET and reaction time on depolymerization were investigated.

3.1.1. Effect of the Weight Ratio of Methanol to PET on Depolymerization. The effect of the weight ratio of methanol to PET on depolymerization at 270 °C and 1000 rpm for the reaction time of 40 min is summarized in Figure 2. The result

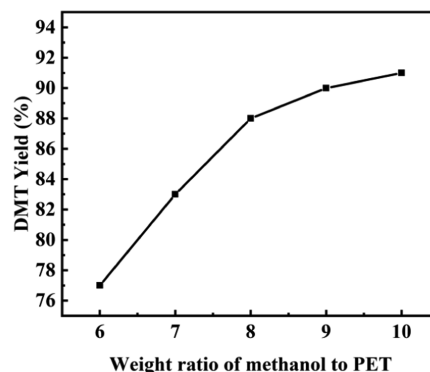


Figure 2. Effect of the weight ratio of methanol to PET on the yield of DMT. Fixed conditions: 270 °C, 1000 rpm, and reaction time of 40 min.

indicates that the yield of DMT increases gradually with the weight ratio of methanol to PET. When the weight ratio of methanol to PET is 10, the yield of DMT is the highest, about 92%. Although the yield of DMT is higher when the weight ratio of methanol to PET is higher, more solvent, methanol, is required at this time. Thus, the optimal weight ratio of methanol to PET is 6:1.

3.1.2. Effect of the Reaction Time on Depolymerization. The effect of the reaction time on depolymerization at 270 °C and 1000 rpm for the weight ratio of methanol to PET of 6 is summarized in Figure 3. The result indicates that the yield of DMT increases gradually with the reaction time. When the reaction time is 20 min, the yield of DMT is very low, only about 20%. When the reaction time is 60 min, the yield of DMT can reach 80% and tends to be stable. When the reaction time is 40 min, the yield of DMT is almost the same as that at 60 min. Considering comprehensively, the optimal reaction time is 40 min.

It is shown from the above-mentioned depolymerization results of PET on the weight ratio of methanol to PET and the

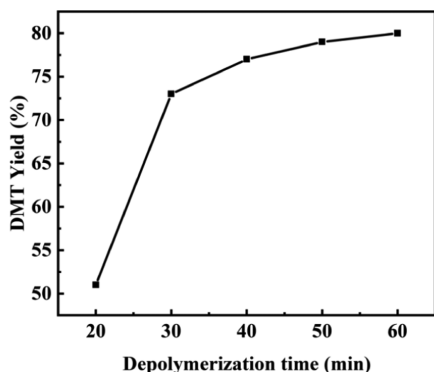


Figure 3. Effect of the reaction time on the yield of DMT. Fixed conditions: 270 °C, 1000 rpm, and weight ratio of methanol to PET of 6:1.

reaction time and the comprehensive study of the literature that the optimal depolymerization conditions are a temperature of 270 °C, a weight ratio (methanol to PET) of 6, a reaction time of 40 min, and a rotational speed of 1000 rpm.

To verify whether the addition of CO₂ can improve the depolymerization efficiency of supercritical methanol, under the optimal depolymerization conditions, the effect of the initial pressure of CO₂ on the depolymerization of PET was also investigated. The results of depolymerization are summarized in Figure 4. The results indicate that the yield

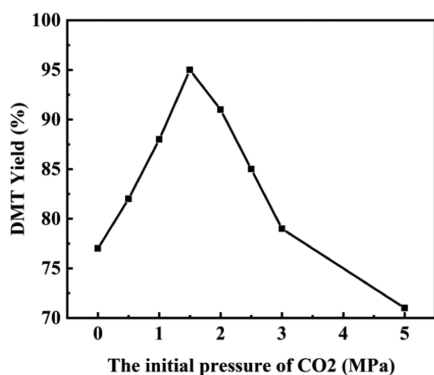


Figure 4. Effect of the initial pressure of CO₂ on the yield of DMT. Fixed conditions: 270 °C, 1000 rpm, 40 min, and weight ratio of methanol to PET of 6:1.

of DMT increases gradually with the initial pressure of CO₂. When the initial pressure of CO₂ reaches 1.5 MPa, the yield of DMT reaches the maximum, which is about 95%. When the initial pressure of CO₂ exceeds 1.5 MPa, the yield of DMT decreases gradually with the increase in the initial pressure of CO₂, and at 5 MPa, the yield of DMT is lower than that without CO₂. Therefore, the process of adding 1.5 MPa CO₂ to strengthen the supercritical methanolysis of PET was selected for investigation.

3.2. Investigation of CO₂-Enhanced Alcoholysis.

3.2.1. Effect of the Reaction Time on Depolymerization. The effect of the reaction time on depolymerization at 270 °C at the initial pressure of CO₂ of 1.5 MPa and 1000 rpm for the weight ratio of methanol to PET of 6 is summarized in Figure 5. The result indicates that the yield of DMT increases gradually with the reaction time. When the reaction time is 20 min, the yield of DMT without CO₂ and with 1.5 MPa CO₂ is about 25%, and the yield of DMT can be increased to about

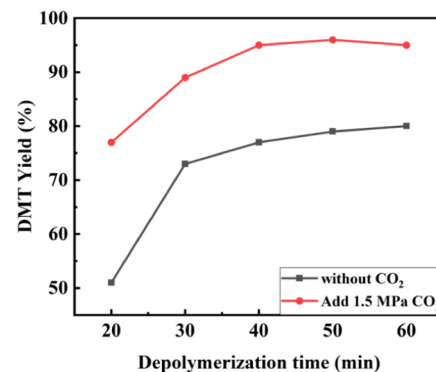


Figure 5. Effect of the reaction time on the yield of DMT with 1.5 MPa CO₂. Fixed conditions: 270 °C, 1000 rpm, and weight ratio of methanol to PET of 6:1.

75% by adding 1.5 MPa CO₂. When the reaction time is 40 min, the yield of DMT without CO₂ and with 1.5 MPa CO₂ is about 20%. When the reaction time reaches 40 min, the yield of DMT is basically stable.

3.2.2. Effect of the Weight Ratio of Methanol to PET on Depolymerization. The effect of the weight ratio of methanol to PET on depolymerization at 270 °C at the initial pressure of CO₂ of 1.5 MPa and 1000 rpm for the reaction time of 40 min is summarized in Figure 6. When CO₂ is not added, the yield

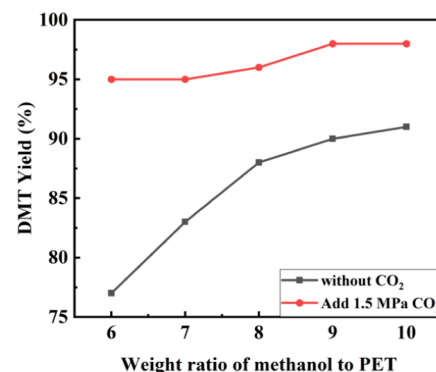


Figure 6. Effect of the weight ratio of methanol to PET on the yield of DMT with 1.5 MPa CO₂. Fixed conditions: 270 °C, 1000 rpm, and 40 min.

of DMT increases gradually with the weight ratio of methanol to PET. When adding 1.5 MPa CO₂, the yield of DMT is basically unchanged and is about 95% with the weight ratio of methanol to PET. The result indicates that the addition of CO₂ can not only effectively improve the supercritical methanolysis efficiency but also save the amount of the solvent.

3.2.3. Effect of Reaction Temperature on Depolymerization. The effect of reaction temperature on depolymerization at 40 min at the initial pressure of CO₂ of 1.5 MPa and 1000 rpm for the weight ratio of methanol to PET of 6 is summarized in Figure 7. The result indicates that the yield of DMT increases gradually with the reaction temperature. Under the same conditions, adding 1.5 MPa CO₂ can effectively improve the yield of DMT. Near the methanol critical temperature (240 °C), the addition of 1.5 MPa CO₂ can increase the yield of DMT from 50% to about 80%. When the reaction temperature is 270 °C, the yield of DMT can be increased to about 95% by adding 1.5 MPa CO₂. With the

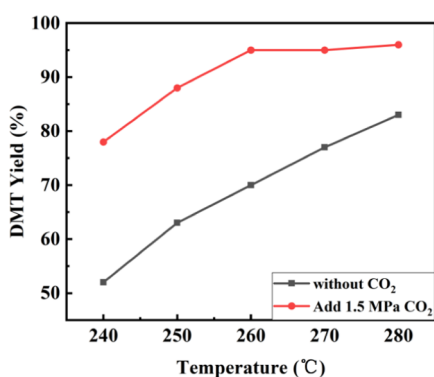


Figure 7. Effect of the reaction temperature on the yield of DMT with 1.5 MPa CO₂. Fixed conditions: 270 °C, 1000 rpm, and weight ratio of methanol to PET of 6:1.

increase of temperature, the yield of DMT without CO₂ and with 1.5 MPa CO₂ is close to each other.

3.2.4. Comparison of the DMT Yield with the Literature. Due to different experimental conditions, there are also differences in the DMT yield between this paper and the literature^{21–23} without CO₂. The DMT yield obtained by CO₂-enhanced alcoholysis was compared with that in the literature. The results are shown in Table 1. Sako²¹ et al. proposed the

Table 1. Comparison between the Yield of DMT Obtained by CO₂-Enhanced Alcoholysis and That in the Literature

literature	temperature (°C)	time (min)	weight ratio	CO ₂ pressure (MPa)	DMT yield (%)
Sako ²¹	300	30		0	80
Kumazawa ²²	310	40	10	0	98
Goto ²³	300	60	5	0	95
this paper	270	40	6	1.5	95

application of supercritical methanol to depolymerize PET into DMT and ethylene glycol and obtained a yield of DMT of 80% during 30 min at 300 °C. Kumazawa²² et al. investigated the effect of supercritical methanol alcoholysis of PET into DMT at 280–310 °C, the methanol to PET ratio (weight) of 10, and the reaction time of 10–70 min. It was found that when the temperature was 310 °C and the reaction time was 40 min, the yield of DMT could reach up to 98% and remained basically stable. Goto²³ et al. proposed the supercritical methanol alcoholysis of PET to DMT and ethylene glycol at 300 °C and 14.7 MPa; the ratio (weight) of methanol to PET was 5, and the reaction time was 60 min. They obtained a yield of DMT of 95%. To sum up, the addition of CO₂ strengthens the decomposition of PET in methanol, which could achieve a higher DMT yield under relatively mild experimental conditions.

3.3. Investigation of the Reaction Mechanism of CO₂-Enhanced Supercritical Methanolysis. To understand the role of CO₂ in supercritical methanolysis of PET better, the comparative experiment of N₂ was carried out. The yield of DMT at 270 °C, 40 min, the initial pressure of N₂ of 1.5 MPa, and 1000 rpm for the weight ratio of methanol to PET of 6 is summarized in Figure 8. The result indicates that the addition of 1.5 MPa N₂ cannot improve the yield of DMT, which is basically the same as that of pure methanol. Both are far less than that of 1.5 MPa CO₂. According to “the solubility of various gases in methanol” in the literature,²⁹ the solubility of

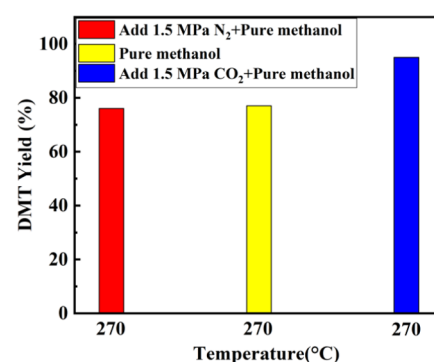


Figure 8. Comparison of CO₂/N₂-strengthened alcoholysis. Fixed conditions: 270 °C, 1000 rpm, and weight ratio of methanol to PET of 6:1.

N₂ in methanol at 270 °C is 0.278648×10^{-3} , which is much less than that of CO₂ in methanol at the same temperature. The solubility of N₂ in methanol is very small. Therefore, the addition of N₂ did not help strengthen the supercritical methanol decomposition of PET.

The reaction mechanism of strengthening the supercritical methanolysis of PET by adding CO₂ is investigated. The *p*-*x*,*y* phase diagram of CO₂ and methanol mixing at a reaction temperature of 270 °C is depicted in Figure 9. The result

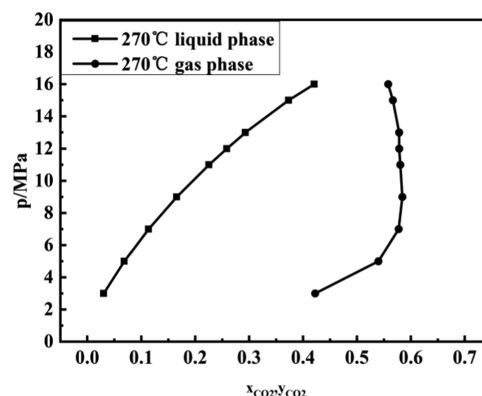


Figure 9. *p*-*x*,*y* diagram at a reaction temperature of 270 °C.

indicates that at this reaction temperature, with the increase of initial pressure, the final pressure of the reaction will also increase. At this time, the solubility of CO₂ in methanol will increase (*x*_{CO₂} will increase), the solubility of CO₂ dissolved in methanol in PET will also increase, greatly increasing the chance of contact with PET, as the CO₂ inserted between PET molecular chains weakens the stress between molecular chains and increases the mobility in PET molecular chains, the crystallinity of PET gradually decreases, which is more conducive to the depolymerization of PET in supercritical methanol, and the yield of DMT is higher. However, more CO₂ is not always better; when the initial pressure of CO₂ is 1.5 MPa and the final pressure is 12 MPa, the corresponding *x*_{CO₂} and *y*_{CO₂} are 0.26 and 0.58, respectively, and *y*_{CH₃OH} is 0.42. When the initial pressure of CO₂ is 5 MPa and the final pressure is 16 MPa, the corresponding *x*_{CO₂} and *y*_{CO₂} are 0.42 and 0.55, respectively, and *y*_{CH₃OH} is 0.45. Therefore, when the addition amount of CO₂ exceeds 1.5 MPa, the content of methanol in the gas phase will be more and more, and CO₂

dissolves too much methanol vapor as a solvent, resulting in insufficient methanol for the methanolysis reaction, so the yield of DMT will decrease.

4. CONCLUSIONS

In this paper, an experimental device for supercritical methanol alcoholysis of PET was established, the conditions affecting the mass ratio of methanol to PET and reaction time were investigated, and the influence of DMT yield assisted by CO₂ was explored. The main conclusions were as follows:

- (1) The optimal reaction conditions of the supercritical methanol alcoholysis of PET were obtained: temperature of 270 °C, mass ratio of methanol to PET of 6:1, reaction time of 40 min, and rotational speed of 1000 rpm.
- (2) The addition of CO₂ has a strengthening effect on the supercritical methanol alcoholysis of PET, which can maximize the yield of DMT (DMT yield is up to 95%). Adding 1.5 MPa CO₂ to enhance the supercritical methanol alcoholysis of PET can effectively reduce the reaction temperature, the amount of the solvent methanol, and the energy consumption of the reaction.
- (3) The insertion of CO₂ between PET molecular chains is the cause of accelerated depolymerization of PET in supercritical methanol. Such insertion weakens the interchain force and increases mobility in PET molecular chains, and the crystallinity of PET decreases gradually.

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Notes

The authors declare no competing financial interest.

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